

Problem 04: Differential Calculus

বিশদ ব্যাখ্যা (বাংলা) এবং ইঞ্জিনিয়ারিং স্ট্যান্ডার্ড নোটস (English Color-Coded Callouts)

GIVEN MATHEMATICAL FUNCTION

$$y = 7 \log x - 5 \log x + 8 \cos x$$

Target: Find the first derivative with respect to x , denoted as $\frac{dy}{dx}$ or y' .

 ENGLISH NOTE — PRE-DIFFERENTIATION ALGEBRAIC SIMPLIFICATION

In competitive engineering examinations, always check for **like terms (সদৃশ পদ)** before applying differentiation operators:

$$7 \log x - 5 \log x = (7 - 5) \log x = 2 \log x$$

Thus, the simplified objective function is:

$$y = 2 \log x + 8 \cos x$$

This reduces the number of operations and minimizes potential algebraic arithmetic errors under exam pressure.

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

ফাংশন পদ	ধ্রুবক সহগ নিয়ম (Constant Multiple)	প্রমিত অন্তরীকরণ সূত্র (Standard Derivative)	ফলাফল (Result)
$2 \log x$	$\frac{d}{dx}[c \cdot f(x)] = c \frac{d}{dx}[f(x)]$	$\frac{d}{dx}(\log x) = \frac{1}{x}$	$2 \cdot \frac{1}{x} = \frac{2}{x}$
$8 \cos x$	$\frac{d}{dx}[c \cdot g(x)] = c \frac{d}{dx}[g(x)]$	$\frac{d}{dx}(\cos x) = -\sin x$	$8(-\sin x) = -8 \sin x$

 ENGLISH NOTE — CRITICAL BUET ADMISSION TRAPS

Trap 1 (Trigonometric Negative Sign Omission):

Remember: $\frac{d}{dx}(\sin x) = +\cos x$, but $\frac{d}{dx}(\cos x) = -\sin x$. Missing the minus sign and writing $+8 \sin x$ will result in immediate penalty.

Trap 2 (Term-by-term Expansion without simplification):

Writing $\frac{dy}{dx} = \frac{7}{x} - \frac{5}{x} - 8 \sin x$ is mathematically valid, but leaving it unsimplified without combining to $\frac{2}{x}$ may lose presentation marks.

২. ধাপে ধাপে পূর্ণাঙ্গ লিখিত সমাধান (Step-by-Step Solution)

Step 01 সদৃশ পদ যোগ-বিয়োগ করে সমীকরণ সরলীকরণ:

$$y = 7 \log x - 5 \log x + 8 \cos x$$

$$y = 2 \log x + 8 \cos x$$

ব্যাখ্যা: $7 \log x$ থেকে $5 \log x$ বিয়োগ করে $2 \log x$ লেখা হলো।

Step 02 উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর নেওয়া:

$$\frac{dy}{dx} = \frac{d}{dx} (2 \log x + 8 \cos x)$$
$$\frac{dy}{dx} = 2 \frac{d}{dx} (\log x) + 8 \frac{d}{dx} (\cos x)$$

ব্যাখ্যা: যোগের নিয়ম ও ধ্রুবক সহগ নিয়ম (Constant Multiple Rule) প্রয়োগ করে পদগুলোকে আলাদা করা হলো।

Step 03 প্রমিত সূত্র বসিয়ে চূড়ান্ত মান নির্ণয়:

$$\frac{dy}{dx} = 2 \left(\frac{1}{x} \right) + 8(-\sin x)$$
$$\frac{dy}{dx} = \frac{2}{x} - 8 \sin x$$

FINAL ANSWER

$$\frac{dy}{dx} = \frac{2}{x} - 8 \sin x \quad \text{or} \quad \frac{2(1 - 4x \sin x)}{x}$$

 ENGLISH NOTE — DOMAIN & ANALYTIC CONTINUITY (ENGINEERING RIGOR)

- **Domain of $y(x)$:** $\log x$ is defined only for $x > 0$, while $\cos x$ is entire on $\mathbb{R}$. Thus, $\text{Dom}(y) = (0, \infty)$.
- **Domain of $\frac{dy}{dx}$:** The derivative has x in denominator, so $\text{Dom}\left(\frac{dy}{dx}\right) = (0, \infty)$.
- **Limit behavior:** As $x \rightarrow 0^+$, $\frac{2}{x} \rightarrow +\infty$, meaning the function has a vertical asymptote as it approaches the y-axis from the right.

 ENGLISH NOTE — SINGLE FRACTION MCQ REPRESENTATIONS

In BUET admission MCQs, options are frequently simplified under a common denominator:

$$\frac{dy}{dx} = \frac{2}{x} - 8 \sin x = \frac{2 - 8x \sin x}{x} = \frac{2(1 - 4x \sin x)}{x}$$

 ENGLISH NOTE — MARKS DISTRIBUTION CHECKLIST

Simplification / Setup (1 M) $\rightarrow$ Evaluation of $\frac{d}{dx}(\log x)$ (1.5 M) $\rightarrow$ Evaluation of $\frac{d}{dx}(\cos x)$ (1.5 M) $\rightarrow$ Final Result (1 M).